

Highlights

Hybrid machine learning and dynamical forecasts for subseasonal temperature prediction at high-Andean stations: a blind benchmark with foundation models and conformal calibration

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- Ridge–CFSv2 hybrid ties CFSv2 at week 1 and outperforms it at weeks 2–4
- Models frozen before a two-year blind test; block bootstrap, Holm correction
- Raw Chronos is over-confident; absolute-temperature context restores spread
- Conformal calibration lifts ensemble 90% coverage from 0.63–0.75 to ~0.90
- ENSO/MJO predictors add measurable skill at weeks 2–4 at five Andean stations

Hybrid machine learning and dynamical forecasts for subseasonal temperature prediction at high-Andean stations: a blind benchmark with foundation models and conformal calibration

Moisés Evangelista Gamarra^{a,*}, Jerremi Aron Chancan Labajos^a

^a*Servicio Nacional de Adiestramiento en Trabajo Industrial, Perú*

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ABSTRACT

Temperature forecasts one to four weeks ahead support frost warnings and agricultural planning in the tropical Andes, where machine learning has rarely been tested at subseasonal leads. We present an artificial intelligence forecasting system for weekly mean temperature anomalies at five Peruvian stations between 2,421 and 4,475 m, evaluated under a protocol in which every model, hyperparameter and the primary system were frozen before two blind years (July 2022 to June 2024) were scored. The system combines quantile forecasts from ridge regression, gradient boosting, a long short-term memory network and the Chronos time-series foundation model, trained on leakage-free anomalies with local and large-scale (El Niño–Southern Oscillation, Madden–Julian Oscillation) predictors, and is compared with the bias-corrected Climate Forecast System version 2 (CFSv2). The ensemble improves on climatology at every horizon (continuous ranked probability skill score 0.30, 0.30 and 0.35 at weeks 1, 2 and 3–4), but its 90% intervals cover only 0.63–0.75 of the outcomes; a conformal calibration selected on development years brings the coverage to 0.87–0.90. A ridge model that uses the CFSv2 forecast as a predictor reaches skill scores of 0.42, 0.38 and 0.41: it matches CFSv2 at week 1 and improves on it at weeks 2 and 3–4. Unadapted Chronos is over-confident (coverage 0.22–0.31) until it receives absolute-temperature context. These results, partly from analyses declared after the blind scores were seen, indicate that calibrated subseasonal guidance is attainable at high-elevation stations when statistical learning and dynamical forecasts are combined.

1. Introduction

Air temperature in the high Andes changes over a few kilometres of distance and a few hundred metres of elevation, and its week-to-week anomalies decide whether a frost damages a potato crop, whether pastures recover after a cold spell, or whether irrigation can be postponed (Condori et al., 2014; López-Moreno et al., 2015; Mountain Research Initiative EDW Working Group, 2015). Forecasts between two and four weeks ahead fall in the subseasonal range, which has long been called a predictability gap: too far for the atmospheric initial conditions to dominate and too close for the slowly varying boundary conditions that drive seasonal forecasts (Vitart et al., 2017; Robertson et al., 2015; Mariotti et al., 2018). Operational subseasonal-to-seasonal (S2S) systems have narrowed that gap over the last decade, and their skill is known to depend on windows of opportunity such as the state of the El Niño–Southern Oscillation (ENSO) and the Madden–Julian Oscillation (MJO) (Mariotti et al., 2020; Domeisen et al., 2022; White et al., 2022). Their grid cells, however, do not resolve the stations at which decisions are taken, and their raw output needs statistical correction before it is useful at a point (Monhart et al., 2018; Robertson et al., 2023).

Machine learning has become a natural candidate for that correction and, more ambitiously, for forecasting the anomalies directly. Gradient-boosted trees, regularised linear models and neural networks have been benchmarked against dynamical models for subseasonal temperature over the contiguous United States (Hwang et al., 2019; He et al., 2021; Mouatadid et al., 2023b) and globally (Weyn et al., 2021; Chen et al., 2024), and probabilistic neural post-processing improves ensemble forecasts (Rasp and Lerch, 2018; Scheuerer et al., 2020). Three aspects of this literature limit what it says about high-mountain stations. First, most evaluations use reanalysis grids in the mid-latitudes, where ENSO and the MJO act differently from the tropical Andes (Garreaud, 2009; Sulca et al., 2017). Second, few studies freeze the model choice before scoring a blind period, although temporal cross-validation without such a separation can

*Corresponding author

✉ mevangelistag@senati.pe (M.E. Gamarra); 1520481@senati.pe (J.A.C. Labajos)

ORCID(s): 0009-0002-5382-1390 (M.E. Gamarra); 0009-0005-6789-076X (J.A.C. Labajos)

overstate skill (Tashman, 2000; Bergmeir and Benítez, 2012; Roberts et al., 2017). Third, time-series foundation models such as Chronos (Ansari et al., 2024) are now offered as zero-shot forecasters, but how their predictive distributions behave on a physical target with strong seasonality has not been examined in this setting; nor has the value of combining station-based learning with an operational dynamical forecast been quantified for the Andes.

This paper addresses these points with a forecasting system for weekly mean temperature anomalies at five Peruvian stations between 2,421 and 4,475 m and an evaluation designed to be hard to over-interpret. Its contributions are:

- **A blind evaluation protocol for subseasonal machine learning.** All models, hyperparameters and the primary ensemble were fixed on three development years and frozen before two blind years were scored; every later analysis is declared as a dated amendment, and every comparison is tested with a moving-block bootstrap and Holm's correction.
- **A hybrid dynamical–statistical model.** A ridge regression that uses the bias-corrected forecast of the Climate Forecast System version 2 (CFSv2) as one more predictor, evaluated against a control fitted on exactly the same rows without that predictor, so that its gain can be attributed.
- **A diagnosis of a time-series foundation model on a physical target.** We show why Chronos produces intervals that are too narrow for window-mean anomalies and how an absolute-temperature context, or the Chronos-Bolt variant, changes the balance between calibration and sharpness.
- **Calibrated probabilistic forecasts.** A conformal calibration of the ensemble, selected on development years only, which corrects the over-confidence of the combined forecast.

Section 2 reviews the related work. Section 3 describes the stations and predictors, Section 4 the forecasting system and its evaluation, and Section 5 the blind results. Section 6 discusses them against the literature and states the limitations, and Section 7 concludes.

2. Related work

2.1. Subseasonal prediction and windows of opportunity

The S2S prediction project established a common framework of weekly and fortnightly targets, reforecasts and verification for the one-to-eight-week range (Vitart et al., 2017; Vitart and Robertson, 2018). Skill at these leads is intermittent: it depends on whether the initial state contains a slowly evolving signal, such as an active MJO, a developing ENSO event or persistent soil-moisture anomalies (Mariotti et al., 2018, 2020; Domeisen et al., 2022). The MJO modulates tropical convection and, through it, temperature and rainfall far from its centre of action (Wheeler and Hendon, 2004; Kiladis et al., 2014), and the reforecasts of the Subseasonal Experiment showed how strongly verification depends on these states (Pegion et al., 2019). The practical response has been to report skill conditioned on the state at issuance, a practice we follow.

2.2. Machine learning for subseasonal temperature

Statistical baselines remain strong at these leads: damped persistence and climatology are hard to beat beyond two weeks (Hwang et al., 2019; DelSole and Tippett, 2014), which is why both are used as references throughout. Gradient-boosted trees (Ke et al., 2017) and regularised regressions on large-scale indices compete with dynamical models for weeks 3–4 over North America (He et al., 2021; Mouatadid et al., 2023b), and deep architectures such as recurrent networks (Hochreiter and Schmidhuber, 1997), temporal fusion transformers (Lim et al., 2021) or global convolutional models (Weyn et al., 2021; Chen et al., 2024) extend the approach. For hydrological forecasting at many gauges, a single recurrent network trained on all basins transfers to ungauged ones (Kratzert et al., 2019b,a), a result that motivates our leave-one-station-out test. Reviews of the field note that gains over simple models are often smaller than architectural complexity suggests (Petropoulos et al., 2022; Slater et al., 2023).

2.3. Time-series foundation models

Foundation models pretrained on large collections of time series now produce probabilistic forecasts without task-specific training. Chronos tokenises scaled values and samples future trajectories from a language-model architecture (Ansari et al., 2024); related work explores foundation models and large language models for forecasting more broadly (Liang et al., 2024; Tan et al., 2024). Their benchmarks emphasise short-horizon accuracy on heterogeneous datasets; the calibration of window-averaged quantities derived from their sample paths, which is what a subseasonal user needs, has received little attention.

2.4. Post-processing and hybrids with dynamical models

Raw dynamical forecasts at subseasonal leads are biased and under-dispersive at stations, and simple local corrections recover much of their value (Monhart et al., 2018). CFSv2 is a widely used coupled system whose reforecasts are openly available (Saha et al., 2014). Statistical post-processing ranges from regression on the ensemble mean to ensemble model output statistics (EMOS) (Gneiting et al., 2005) and neural networks (Rasp and Lerch, 2018); machine-learning models that take dynamical forecasts as inputs or correct their biases have improved subseasonal temperature over North America (He et al., 2021; Mouatadid et al., 2023a), and hybrid forecasting that blends climate predictions with machine learning is an active field (Slater et al., 2023). Whether such a hybrid adds value over both of its components at tropical mountain stations is one of the questions addressed here.

2.5. Probabilistic verification and calibration

A probabilistic forecast should be as sharp as possible subject to being calibrated (Gneiting et al., 2007). The continuous ranked probability score (CRPS) is the standard proper score for continuous variables (Hersbach, 2000; Gneiting and Raftery, 2007), and the probability integral transform (PIT) and interval coverage diagnose calibration (Hamill, 2001; Gneiting and Ranjan, 2011). Combining forecasts by averaging their quantiles (Vincentization) tends to produce sharper but possibly over-confident distributions, whereas linear pooling widens them (Genest, 1992; Lichtendahl et al., 2013). Conformal prediction offers finite-sample coverage guarantees for any model and has been extended to quantile regression (Romano et al., 2019; Angelopoulos and Bates, 2023); we use it as one of the candidate calibrations of the ensemble.

2.6. Temperature variability in the tropical Andes

Interannual temperature variability in the tropical Andes is dominated by ENSO, with warm anomalies during El Niño and a modulation by the regional coastal and central-Pacific modes (Vuille et al., 2000; Vuille and Bradley, 2000; Garreaud, 2009). The coastal (Niño 1+2) and central-Pacific (Niño 3.4) sectors affect Peru differently, which led to the E and C indices (Takahashi et al., 2011); ENSO teleconnections to Andean temperature and rainfall have been documented for the Altiplano and the western slope (Lavado-Casimiro and Espinoza, 2014; Sulca et al., 2017; Imfeld et al., 2020). Warming is elevation-dependent (Mountain Research Initiative EDW Working Group, 2015), and the observing network is sparse and heterogeneous, which makes quality control an integral part of any station-based study (Hunziker et al., 2017; Condori et al., 2014).

3. Study area and data

3.1. Stations

We use hourly records from five automatic stations of the Peruvian national meteorological service, SENAMHI, published as an open dataset (Servicio Nacional de Meteorología e Hidrología del Perú, 2024b) whose variables are defined in the provider's data dictionary (Servicio Nacional de Meteorología e Hidrología del Perú, 2024a), selected to span the western Andean slope and the Altiplano margin from 2,421 to 4,475 m (Fig. 1; Table 1). Two belong to the global basic observing network (GBON) and three to the regional one (RBON). The common record runs from 1 January 2015 to 30 June 2024 (3,469 days per station). The harmonic climatologies in Fig. 1 b illustrate the gradient: the mean daily temperature drops by about 12 °C from Matucana to Imata, and the amplitude of the annual cycle at the highest station is about three times that at the lowest.

Table 1
Stations used in this study (hourly records, 2015–2024).

Station	Department	Network	Latitude (°)	Longitude (°)	Elevation (m)
Matucana	Lima	GBON	−11.839	−76.378	2,421
San José de Uzuña	Arequipa	RBON	−16.581	−71.328	3,269
Candarave	Tacna	RBON	−17.268	−70.254	3,410
Carania	Lima	GBON	−12.344	−75.872	3,840
Imata	Arequipa	RBON	−15.843	−71.091	4,475

Table 2 lists the variables used, from the station dataset and the external sources described in Section 3.3.

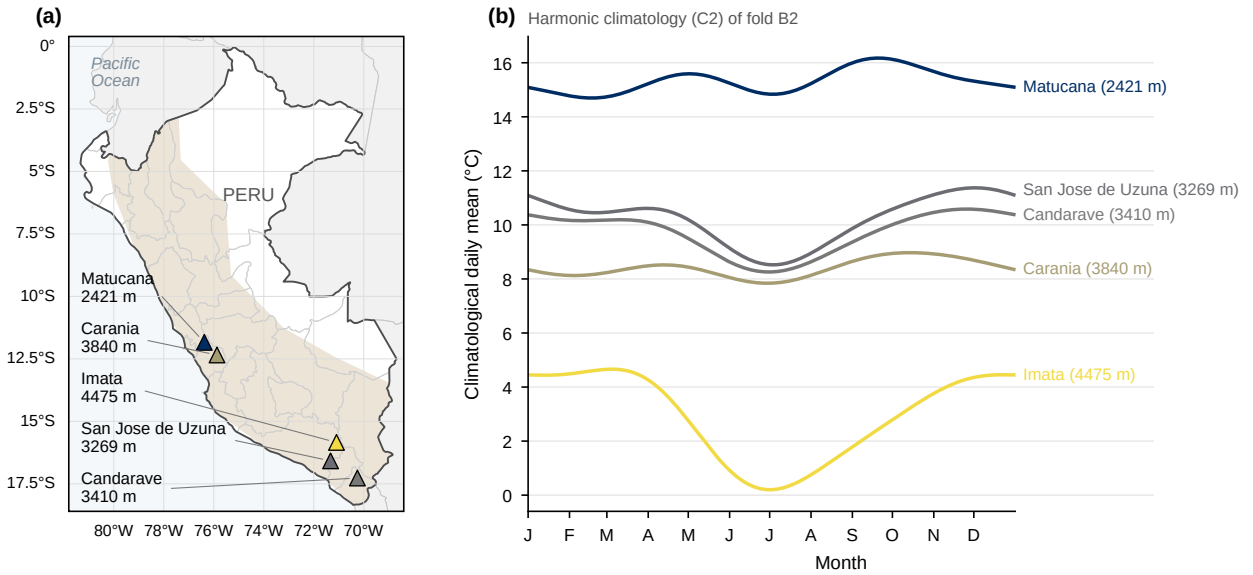


Figure 1: Study area: (a) stations coloured by elevation over the Andes (shaded); (b) climatological daily mean temperature at each station.

Table 2

Variables used in this study.

Variable	Description	Unit	Resolution	Source	Use
TEMP	Air temperature	°C	Hourly	SENAMHI	Target; L
HR	Relative humidity	%	Hourly	SENAMHI	L
PP	Precipitation	mm h ⁻¹	Hourly	SENAMHI	L
ALTITUD	Station elevation	m	Static	SENAMHI	S
LATITUD, LONGITUD	Station coordinates	°	Static	SENAMHI	S
Niño 3.4, Niño 1+2	SST anomaly	°C	Weekly	NOAA CPC	G
ROMI 1, 2, amplitude	MJO index	—	Daily	NOAA PSL	G
E, C	ENSO indices	—	Monthly	IGP	Sensitivity
CFSv2 tmp2m	2 m temperature forecast	°C	6-hourly, 1°	NOAA	CFS_BC; hybrid

Station variable names and definitions follow the provider's data dictionary; TEMP is the hourly mean and PP the hourly total. L, G and S as in the predictor sets of Section 4.3.

3.2. Quality control and daily aggregation

The hourly file contains 416,273 station-hours. We first verified the time convention (the diurnal cycle peaks at 13:00 and bottoms at 06:00 local time, consistent with local civil time), the 24-hour coverage and the absence of duplicated timestamps, and treated empty fields as missing; 13,268 station-hours (3.2%) have no variable at all. Temperature values then went through range, spike and stuck-sensor checks adapted from Hunziker et al. (2017): 428 spikes and 2,114 stuck-sensor hours were flagged. Daily means were computed from the remaining hours, and a day was kept only when enough hours were valid; targets were never imputed.

3.3. Large-scale indices and the dynamical forecast

The large-scale predictors are the weekly Niño 3.4 and Niño 1+2 sea-surface temperature anomalies from the NOAA optimum interpolation analysis (Reynolds et al., 2002; NOAA Climate Prediction Center, 2024), used with a seven-day publication lag, and the two components and amplitude of the real-time outgoing-longwave- radiation MJO index (ROMI) (Kiladis et al., 2014; Wang et al., 2018; NOAA Physical Sciences Laboratory, 2024), used with a one-day lag. The E and C indices (Takahashi et al., 2011; Instituto Geofísico del Perú (IGP), 2026) enter only as a sensitivity analysis. Indices published with a delay longer than the issuance (such as the Oceanic Niño Index) were

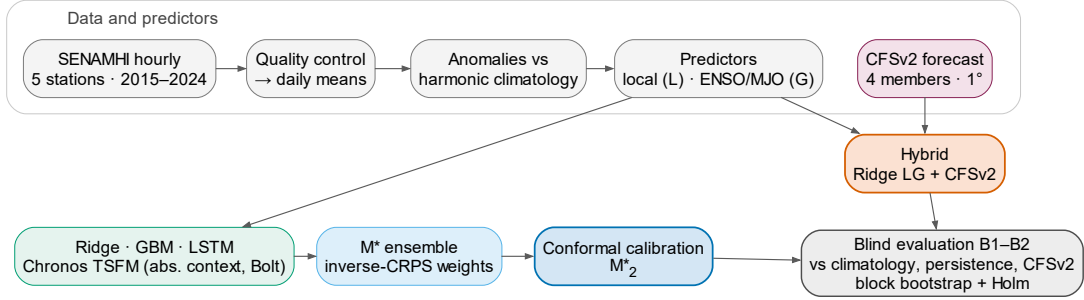


Figure 2: Forecasting system, from station data to blind evaluation.

excluded to avoid look-ahead. The dynamical forecast is the four-member 00 UTC CFSv2 run issued each Monday (Saha et al., 2014; NOAA National Centers for Environmental Prediction, 2024), whose 6-hourly 2 m temperature on the 1° grid was interpolated bilinearly to each station and averaged over each target window; the open archive starts on 31 October 2018.

4. Methodology

4.1. Problem formulation

A forecast is issued every Monday t for each station s and three targets: the mean temperature anomaly over days 1–7 (week 1, W1), days 8–14 (week 2, W2) and days 15–28 (weeks 3–4, W3–4) after issuance. Window means are the standard S2S target because day-to-day weather is unpredictable at these leads while averages retain signal (Vitart et al., 2017; Robertson et al., 2015; Pegion et al., 2019), and the same three windows are used by recent machine-learning benchmarks (Hwang et al., 2019; He et al., 2021; Mouatadid et al., 2023b). Each horizon has its own model (direct strategy), so no forecast is fed back as an input. Models are trained on issuances made every day of the training period, which multiplies the sample size, and evaluated on Mondays only. The complete system is shown in Fig. 2.

4.2. Anomaly decomposition

All models forecast anomalies with respect to a climatology fitted only on each fold’s training days, so that no information from the test period enters the reference. The climatology of station s is a harmonic regression of the daily mean temperature on the day of the year d (Epstein, 1991), Eq. 1:

$$c_s(d) = a_0 + \sum_{k=1}^K \left[a_k \cos\left(\frac{2\pi kd}{P}\right) + b_k \sin\left(\frac{2\pi kd}{P}\right) \right], \quad K = 3, \quad P = 365.25, \quad (1)$$

with $K = 3$ chosen on the development years against $K = 1$ and $K = 5$ by the out-of-sample variance of the monthly mean residual. Three harmonics capture the asymmetric annual cycle of the Altiplano stations, which a single harmonic cannot represent (Vuille and Bradley, 2000; Mountain Research Initiative EDW Working Group, 2015). The target (Eq. 2) is the mean anomaly over the window $\mathcal{W}_h$ of horizon h ,

$$y_{s,t,h} = \frac{1}{|\mathcal{W}_h|} \sum_{\ell \in \mathcal{W}_h} [T_s(t + \ell) - c_s(d(t + \ell))], \quad (2)$$

where T_s is the daily mean temperature. A window counts only when at least 5 of 7 (W1, W2) or 10 of 14 (W3–4) days are valid.

4.3. Predictors

The local predictor set (L) contains lagged daily temperature anomalies (lags 0–6 days) and their 7-, 14- and 30-day means, anomalies of the diurnal temperature range, daily maximum and minimum temperature and relative humidity, log-transformed 7- and 30-day rainfall totals, and the sine and cosine of the target day of the year, all computed from trailing windows that end on the issuance date; persistence of recent anomalies is the main source of skill at week 1 (DelSole and Tippet, 2014). The large-scale set (G) adds the Niño 3.4, Niño 1+2 and ROMI predictors described above, because ENSO and the MJO are the main sources of subseasonal predictability for the region (Lavado-Casimiro and Espinoza, 2014; Sulca et al., 2017; Mariotti et al., 2020). Models with the suffix LG use L and G, models with the suffix L only L, which isolates the contribution of the large-scale indices. The CFSv2 predictor is described in Section 4.6. Table 3 summarises inputs and outputs.

Table 3

Predictor sets and forecasting systems.

System	Inputs	Distribution	Status
Climatology (Clim)	Training targets, ± 15 days	Empirical	Ref.
Damped persistence (Damp)	Last 7-day anomaly	Gauss.	Ref.
CFS_BC	CFSv2 window mean	Gauss.	Ref. (A1)
Ridge L / Ridge LG	L / L + G	Gauss.	Cand.
GBM L / GBM LG	L / L + G, with S	Q19	Cand.
LSTM LG	60-day sequences of L + G, with S	Q19	Cand.
Chronos	Daily anomalies, 512 days	100 paths	Cand.
Chronos (abs.), Chronos-Bolt	Absolute temperature	Paths, Q19	A3
M* (ensemble)	Ridge LG, GBM LG, LSTM LG, Chronos	Q19, Vincentized	Primary
M*_2	M* members, Chronos (abs.)	Q19, conformal	A3
Hybrid	L + G + CFSv2 anomaly	Gauss.	A5

L: local predictors; G: large-scale predictors; S: elevation, latitude and longitude. Gauss.: Gaussian with the training residual SD by quarter; Q19: 19 quantile levels (0.05–0.95). Ref.: reference; Cand.: pre-registered candidate; A3, A5: post hoc amendments.

4.4. Forecasting models

References. Climatology and damped persistence are the references against which skill is measured, because a subseasonal forecast is only useful if it improves on what these simple rules already provide (Murphy, 1993; DelSole and Tippet, 2014). Damped persistence regresses the target on the most recent 7-day anomaly $A_{s,t}^{(7)}$ for each station, horizon and season quarter q (Eq. 3):

$$\hat{y}_{s,t,h} = \beta_{0,shq} + \beta_{1,shq} A_{s,t}^{(7)}, \quad Y \sim \mathcal{N}(\hat{y}, \sigma_{shq}^2), \quad (3)$$

with σ_{shq} the standard deviation of the training residuals of that cell. The bias-corrected CFSv2 forecast (CFS_BC) follows the station-level correction of Monhart et al. (2018): the CFSv2 window mean $F_{s,t,h}$ is converted to an anomaly with respect to its own harmonic climatology $\tilde{c}_{sh}$ ($K = 2$, fitted on the training Mondays), and the observed anomaly is regressed on it,

$$f_{s,t,h} = F_{s,t,h} - \tilde{c}_{sh}(d_{\text{mid}}), \quad \hat{y}_{s,t,h} = \gamma_{0,sh} + \gamma_{1,sh} f_{s,t,h}, \quad (4)$$

with a Gaussian distribution whose standard deviation is that of the training residuals by quarter, as for damped persistence, so that differences in CRPS come from the mean.

Ridge regression. A standardised ridge regression per station and horizon, with the penalty chosen on the last training year separated by the embargo, is the simplest learned model; its predictive distribution is Gaussian with the inner-validation residual standard deviation by quarter.

Gradient boosting. LightGBM (Ke et al., 2017) is pooled over stations per horizon, with elevation, latitude and longitude as static descriptors. One model is fitted per quantile level τ by minimising the pinball loss of quantile regression (Koenker and Bassett, 1978) (Eq. 5),

$$\rho_{\tau}(u) = u(\tau - \mathbb{1}\{u < 0\}), \quad \hat{q}_{\tau} = \arg \min_q \sum_i \rho_{\tau}(y_i - q(x_i)), \quad (5)$$

and quantile crossings are removed by rearrangement, which never increases the quantile loss (Chernozhukov et al., 2010).

Long short-term memory network. An LSTM (Hochreiter and Schmidhuber, 1997) reads 60-day sequences of the LG predictors, together with the static descriptors of the station, and is trained on all stations at once, following the regional training strategy of Kratzert et al. (2019b); it outputs the 19 quantiles with the pinball loss, with early stopping on the last training dates separated by the embargo.

Time-series foundation model. Chronos-T5-small (Ansari et al., 2024) forecasts the daily anomaly series from a 512-day context, and each of its 100 sample paths is averaged over the target window, which gives an empirical distribution of the window mean. After the first blind scores showed intervals that were far too narrow (Section 5.5), amendment A3 added two configurations: Chronos with the absolute temperature as context, the climatology being subtracted afterwards, and Chronos-Bolt, which returns quantiles of weekly and fortnightly block means directly. All Chronos variants are used zero-shot, without fine-tuning. Table 4 lists every hyperparameter.

Table 4

Hyperparameters, frozen before the blind folds.

Component	Setting
Ridge	$\lambda \in \{0.1, 1, 10, 100, 1000\}$
GBM (LightGBM)	300 trees, learning rate 0.03, 15 leaves, ≥ 40 samples per leaf Row and column subsampling 0.8; one model per quantile
LSTM	60-day sequences, 32 units, 1 layer, dropout 0.1 30 epochs, batch 256, learning rate 10^{-3} , patience 5
Chronos	chronos-t5-small, 512-day context, 100 paths, bfloat16
Chronos-Bolt (A3)	chronos-bolt-small, 7- and 14-day block means
Quantile grid	19 levels, 0.05–0.95, rearranged
M*	Vincentization, inverse development-CRPS weights
M* ₂ (A3)	Best of 16 configurations, development coverage 0.85–0.95
Hybrid (A5)	Ridge LG settings, no tuning
Validation	Expanding window from 2015, 28-day embargo, 5 folds
Bootstrap	Blocks of 8 weeks (4 and 13 as sensitivity), $B = 10,000$
Seeds	GBM 101/202, LSTM 505, bootstrap 303; 4 extra seeds (A3)

The ridge penalty is selected per station and horizon on the last training year, after the embargo. Chronos-Bolt levels 0.05 and 0.95 come from a normal fitted to its 0.1–0.9 spread.

4.5. Ensemble and calibration

The primary system M* averages the quantiles of the candidate models (Vincentization) with weights inversely proportional to their mean CRPS on the development folds, per horizon (Eq. 6):

$$q_{\tau}^{M^*} = \sum_m w_{m,h} q_{\tau}^{(m)}, \quad w_{m,h} = \frac{1/\overline{\text{CRPS}}_{m,h}^{\text{dev}}}{\sum_{m'} 1/\overline{\text{CRPS}}_{m',h}^{\text{dev}}}. \quad (6)$$

Combining forecasts generally reduces error relative to the members (Bates and Granger, 1969; Buseti, 2017), and Vincentization keeps the combined distribution sharp (Genest, 1992; Lichtendahl et al., 2013). M* was chosen as primary because its pooled development CRPS (0.3926 °C) was the lowest among the candidates, ahead of Ridge LG (0.3943 °C), LSTM LG (0.3946 °C) and GBM LG (0.4207 °C); the weights were close to equal (0.22–0.27 per member).

Vincentization is known to produce over-confident distributions when the members disagree (Lichtendahl et al., 2013), and the blind coverage of M* confirmed this (Section 5.4). Amendment A3 therefore defined M*₂, chosen on the development folds only among 16 configurations: two member sets (with or without the best Chronos variant), two combinations (Vincentization or linear pool) and four calibrations per horizon (none, a spread factor, EMOS (Gneiting et al., 2005) and conformalised quantiles (Romano et al., 2019; Angelopoulos and Bates, 2023)). The conformal calibration widens each central pair of quantiles ($\tau, 1 - \tau$) by the empirical quantile of the conformity score on the calibration rows (Eq. 7),

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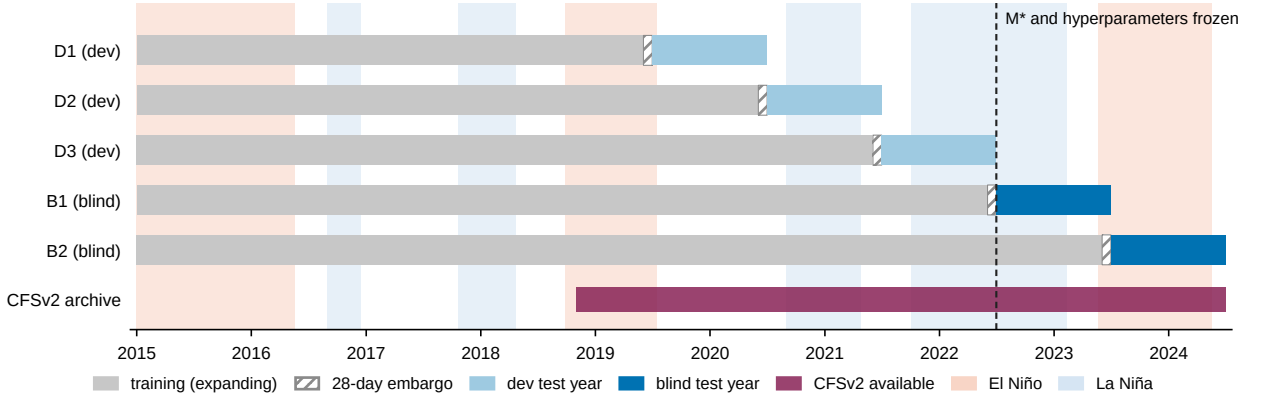


Figure 3: Validation design: expanding training windows, 28-day embargo, development and blind test years, CFSv2 archive and ENSO episodes; models were frozen at the dashed line.

$$E_i = \max(q_{\tau,i} - y_i, y_i - q_{1-\tau,i}), \quad \delta_\tau = \hat{Q}_{[(n+1)(1-2\tau)]/n}(\{E_i\}), \quad \tilde{q}_\tau = q_\tau - \delta_\tau, \quad \tilde{q}_{1-\tau} = q_{1-\tau} + \delta_\tau, \quad (7)$$

leaving the median unchanged. Weights and calibrations were cross-fitted (fitted on two development folds, scored on the third), and the winner was the configuration with the lowest cross-fitted CRPS whose 90% coverage lay in 0.85–0.95 at every horizon. It was refitted on all development years and written to disk before any blind row was scored.

4.6. Hybrid dynamical–statistical model

Amendment A5, written after the blind scores of M* and CFS_BC were known, tests whether the two sources of information are complementary. The hybrid (Eq. 8) adds the CFSv2 anomaly of Eq. 4 to the ridge regression of Ridge LG,

$$\hat{\mu}_{s,t,h} = \beta_0 + \beta_L^\top \mathbf{x}_{s,t}^L + \beta_G^\top \mathbf{x}_t^G + \beta_F f_{s,t,h}, \quad \min_{\beta} \sum_i (y_i - \hat{\mu}_i)^2 + \lambda \|\beta_{-0}\|_2^2, \quad (8)$$

with the same penalty grid, inner validation and residual distribution as Ridge LG. Using a dynamical forecast as a predictor of a statistical model follows earlier subseasonal hybrids (He et al., 2021; Mouatadid et al., 2023a). Because the CFSv2 archive starts in October 2018, the hybrid can only be trained in the blind folds, on about 185 training Mondays per station and horizon, and nothing could be tuned. To separate the effect of the CFSv2 predictor from that of the shorter, weekly training sample, we fitted a control with the same rows and without f ; this control has no direct precedent in the literature and follows from the logic of a paired ablation. We also report a gradient-boosting hybrid and an untrained blend, the equal-weight quantile average of CFS_BC and M*.

4.7. Validation design and leakage control

Figure 3 shows the design. Five one-year test folds, from July to June, are evaluated with an expanding training window that starts on 1 January 2015 (rolling origin; Tashman (2000)). The first three folds (D1–D3, 2019–2022) were used for development; the last two (B1, B2, July 2022 to June 2024) were blind. A 28-day embargo separates the last training target from the first test day, so that no training window overlaps a test window, as required for serially dependent data (Bergmeir and Benítez, 2012; Roberts et al., 2017). The climatology, scalars, penalties, residual distributions and ensemble weights are all estimated on training rows only. Hyperparameters and the primary model were frozen before B1; the blind folds contain 102, 100 and 96 issuance dates at W1, W2 and W3–4 (506, 497 and 478 station forecasts). Because a W3–4 forecast issued in June would have a target in the next fold’s test year, those issuances are not evaluated in either fold. Transfer to stations without history was tested by leave-one-station-out (LOSO) training of the pooled models on the blind folds (Kratzert et al., 2019a). In Fig. 3, El Niño and La Niña episodes are periods of at least 60 days in which the 91-day mean Niño 3.4 anomaly exceeds +0.5 °C or falls below −0.5 °C.

Reproducibility. The pipeline, its configuration and the two notebooks that run it end to end are archived on Zenodo (Evangelista Gamarra and Chancan Labajos, 2026), and the station data are the open dataset described in Section 3. Every number in this paper is read from a machine-readable output of that pipeline.

4.8. Verification metrics

The CRPS is computed from the 19 quantiles as twice the mean pinball loss, which approximates the integral of the quantile score over τ (Gneiting and Raftery, 2007; Hersbach, 2000) (Eq. 9):

$$\text{CRPS}(F, y) \approx \frac{2}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \rho_{\tau}(y - q_{\tau}). \quad (9)$$

Skill against a reference r is reported as a skill score computed from the sums over all stations and issuances, which weights each forecast equally and avoids the instability of averaging per-case ratios (Murphy, 1988; Wilks, 2019) (Eq. 10):

$$\text{CRPSS}_r = 1 - \frac{\sum_i \text{CRPS}_i}{\sum_i \text{CRPS}_i^{(r)}}, \quad \text{MSSS}_r = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \hat{y}_i^{(r)})^2}. \quad (10)$$

Calibration is diagnosed with the 90% interval coverage and the PIT (Hamill, 2001; Gneiting and Ranjan, 2011) (Eq. 11):

$$\text{Cov}_{90} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{q_{0.05,i} \leq y_i \leq q_{0.95,i}\}, \quad u_i = \hat{F}_i(y_i), \quad (11)$$

where a calibrated forecast has $\text{Cov}_{90} \approx 0.90$ and uniform PIT values; coverage in 0.85–0.95 was the pre-registered criterion. We also report the mean absolute error, root-mean-square error, anomaly correlation (ACC), the ranked probability skill score for terciles (RPSS) (Epstein, 1969) and the mean width of the 90% interval as a measure of sharpness.

4.9. Statistical inference and pre-registration

All tests use the blind folds only, with stations pooled. Because the forecasts of consecutive weeks and of different stations are correlated, differences are aggregated per issuance date and resampled by a moving-block bootstrap of 8 weeks (Kunsch, 1989), with the same blocks for every station; the nested ridge comparison uses the Clark–West statistic with a heteroscedasticity- and autocorrelation-consistent variance (Clark and West, 2007; Newey and West, 1987). A one-sided p-value is the share of bootstrap replicates of the centred statistic at least as large as the observed one, and p-values are adjusted by Holm’s procedure within each family of three horizons (Holm, 1979; Wilks, 2016) (Eq. 12):

$$p = \frac{1 + \#\{b : \Delta_b^* - \bar{\Delta} \geq \bar{\Delta}\}}{B + 1}, \quad p_{(j)}^{\text{Holm}} = \max_{i \leq j} \min(1, (m - i + 1) p_{(i)}). \quad (12)$$

The design, the hypotheses H1–H4 and the decision rule were written before any score existed; analyses added after the blind scores were seen are dated amendments, reported as such (Table 5).

5. Results

5.1. Blind skill overview

Figure 4 and Tables 6–8 summarise the blind results. Every learned model improves on climatology at every horizon, and every one except raw Chronos improves on damped persistence. Two patterns stand out. CFS_BC is the best single system at W1 (CRPSS against climatology 0.415, 95% interval 0.32–0.50), but its skill falls to 0.218 (0.14–0.28) at W3–4, the steepest decline of any system; the dynamical forecast is most useful where its initial conditions still matter, as expected from the S2S literature (Vitart et al., 2017; Pegion et al., 2019). The statistical and machine-learning models follow the opposite course: their skill is flat or increases with the horizon, from 0.297 to 0.385 for Ridge LG and from 0.273 to 0.378 for LSTM LG. Window means over 14 days filter out unpredictable daily weather and keep

Table 5

Hypotheses and amendments.

ID	Statement	Status
H1	LG better than L, for Ridge and GBM	Pre-registered
H2	M* better than Damp at W3–4 (MSSS)	Pre-registered
H3	M* skilful against Clim (CRPSS)	Pre-registered
H4	M* better than CFS_BC (CRPS)	Pre-registered (A1)
A2	Calibration diagnostics of M* and Chronos	Post hoc
H5	M* ₂ better than M*, coverage 0.85–0.95	Post hoc (A3)
H6	Hybrid better than CFS_BC, M* and control	Post hoc (A5)

Pre-registered: written before any score existed. Post hoc: written after the blind scores were seen. Amendment A4 only changed the scope of the secondary analyses.

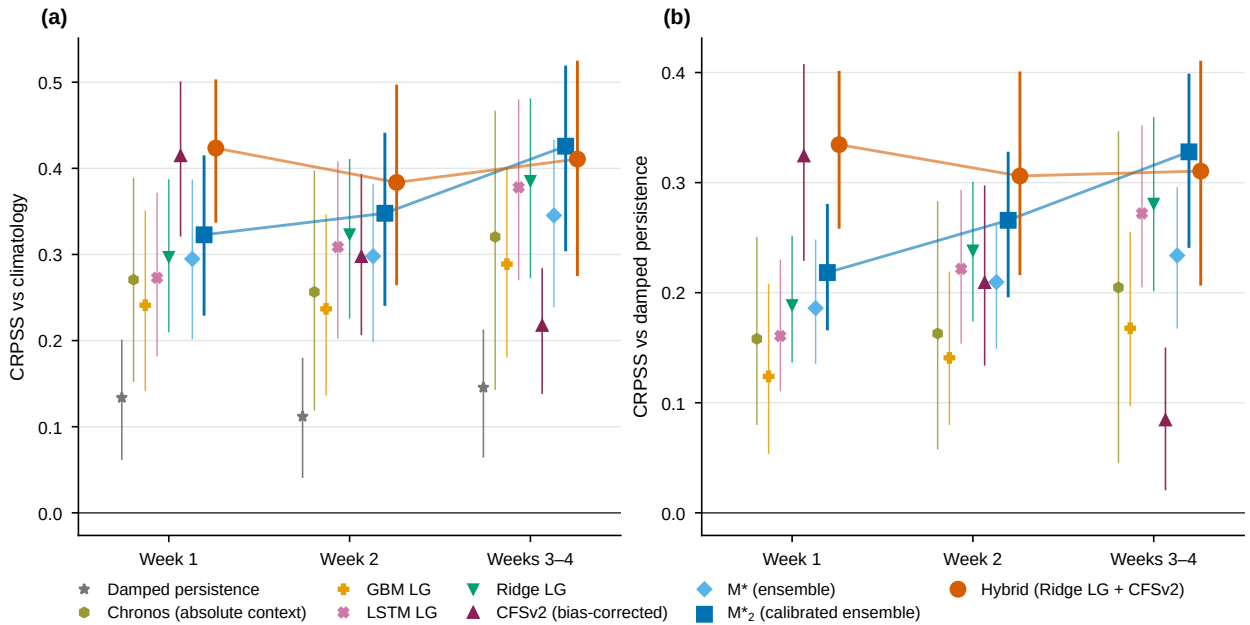


Figure 4: Blind skill by horizon against (a) climatology and (b) damped persistence, with 95% intervals; filled markers: interval above zero.

the slowly varying component driven by persistence and large-scale indices, the mechanism that Mariotti et al. (2018) and Domeisen et al. (2022) identify as the source of subseasonal skill.

The primary ensemble M* has CRPSS of 0.295 (0.20–0.39), 0.298 (0.20–0.38) and 0.345 (0.24–0.43) against climatology, which supports H3 (Holm $p = 0.0003$ at each horizon), and its mean-square skill against damped persistence at W3–4 is 0.409 (one-sided 95% lower bound 0.339; $p = 0.0001$), which supports H2. M* is not, however, better than its best member: Ridge LG reaches 0.297, 0.324 and 0.385, within the sampling uncertainty of M* at every horizon. The deterministic metrics tell the same story (Table 8). At W3–4 Ridge LG has the lowest root-mean-square error (0.612°C) and the highest mean-square skill (0.598), and LSTM LG has the highest anomaly correlation together with M* (0.794 and 0.795).

5.2. Value of the large-scale predictors

The comparison of each model with and without the ENSO and MJO predictors tests H1 (Table 10; Fig. 13). For the ridge regression the gain is clear at every horizon: the Clark–West statistics are 4.10, 4.56 and 4.55 (Holm $p < 10^{-4}$), and the CRPS difference is 0.22, 0.34 and 0.41°C summed over stations per issuance (Holm $p = 0.016$, 0.009 and 0.009). For gradient boosting the large-scale predictors help at W2 and W3–4 (Holm $p = 0.048$ and 0.004 for

Table 6

Blind verification metrics at week 1.

System	MAE	RMSE	ACC	MSSS _{Clim}	CRPS	CRPSS _{Clim}	CRPSS _{Damp}	RPSS	Cov90	Width90
Clim	0.939	1.188	—	0.000	0.705	0.000	−0.154	0.000	0.773	2.851
Persistence	0.872	1.117	0.533	0.115	—	—	—	—	—	—
Damp	0.823	1.038	0.492	0.235	0.611	0.134	0.000	0.157	0.814	2.721
CFS_BC	0.552	0.711	0.788	0.641	0.413	0.415	0.324	0.423	0.852	2.006
Ridge L	0.713	0.918	0.668	0.403	0.540	0.235	0.117	0.293	0.802	2.349
Ridge LG	0.656	0.848	0.709	0.490	0.496	0.297	0.189	0.330	0.812	2.263
GBM L	0.754	0.969	0.596	0.334	0.549	0.222	0.101	0.263	0.777	2.211
GBM LG	0.708	0.918	0.617	0.402	0.535	0.241	0.124	0.264	0.769	2.040
LSTM LG	0.687	0.878	0.693	0.454	0.513	0.273	0.161	0.280	0.838	2.528
Chronos	0.680	0.850	0.687	0.488	0.617	0.126	−0.009	0.206	0.219	0.488
M*	0.651	0.840	0.727	0.500	0.497	0.295	0.186	0.339	0.753	1.889
Chronos (abs.)	0.672	0.845	0.689	0.493	0.514	0.271	0.158	0.306	0.739	2.028
Chronos-Bolt	0.749	0.960	0.586	0.347	0.555	0.213	0.091	0.246	0.895	3.054
M* ₂	—	—	—	—	0.478	0.323	0.218	—	0.903	—
Hybrid	—	—	—	—	0.407	0.424	0.335	—	0.836	—

Folds B1–B2, stations pooled. MAE, RMSE and Width90 in °C; MSSS and CRPSS are skill scores against climatology (Clim) or damped persistence (Damp); Cov90: 90% interval coverage (nominal 0.90). Bold: best pre-registered system (Clim to M*); for Cov90, closest to 0.90. The last four rows are post hoc systems, for which only probabilistic scores were computed.

Table 7

Blind verification metrics at week 2.

System	MAE	RMSE	ACC	MSSS _{Clim}	CRPS	CRPSS _{Clim}	CRPSS _{Damp}	RPSS	Cov90	Width90
Clim	0.846	1.057	—	0.000	0.641	0.000	−0.126	0.000	0.716	2.361
Persistence	0.817	1.050	0.545	0.014	—	—	—	—	—	—
Damp	0.754	0.941	0.467	0.208	0.570	0.112	0.000	0.122	0.746	2.255
CFS_BC	0.609	0.754	0.702	0.491	0.450	0.298	0.209	0.322	0.805	1.977
Ridge L	0.652	0.833	0.650	0.379	0.502	0.217	0.118	0.287	0.708	1.911
Ridge LG	0.569	0.731	0.733	0.521	0.434	0.324	0.238	0.389	0.787	1.790
GBM L	0.737	0.924	0.502	0.235	0.542	0.155	0.049	0.204	0.654	1.704
GBM LG	0.633	0.813	0.623	0.409	0.489	0.237	0.141	0.299	0.656	1.544
LSTM LG	0.594	0.749	0.737	0.498	0.443	0.309	0.222	0.353	0.799	2.070
Chronos	0.661	0.820	0.641	0.398	0.585	0.087	−0.028	0.110	0.290	0.606
M*	0.581	0.742	0.749	0.506	0.450	0.298	0.210	0.378	0.686	1.545
Chronos (abs.)	0.637	0.789	0.659	0.443	0.477	0.256	0.163	0.297	0.841	2.378
Chronos-Bolt	0.662	0.848	0.617	0.356	0.499	0.222	0.124	0.291	0.954	3.345
M* ₂	—	—	—	—	0.418	0.348	0.266	—	0.887	—
Hybrid	—	—	—	—	0.395	0.384	0.306	—	0.843	—

Conventions as for week 1.

CRPS) but not at W1 ($p = 0.26$). The pattern, no gain from remote indices at week 1 and an increasing gain after it, agrees with the windows-of-opportunity view in which persistence dominates the first week and slowly varying modes the following ones (Mariotti et al., 2020; Domeisen et al., 2022). Replacing the Niño indices by the E and C indices of Takahashi et al. (2011) changes the CRPSS of Ridge LG and GBM LG by at most 0.02 (sensitivity runs, not tabulated), so the result does not depend on how ENSO is described.

5.3. Against the dynamical model and the hybrid

The pre-registered comparison with CFS_BC (H4) is mixed. M* is clearly worse at W1 (Δ CRPS -0.419), indistinguishable at W2 (0.001; Holm $p = 0.69$) and better at W3–4 (0.383; Holm $p = 0.012$). The crossover between a dynamical forecast that is strong early and a statistical one that is stronger later reproduces, at stations and at a tropical

Table 8

Blind verification metrics at weeks 3–4.

System	MAE	RMSE	ACC	MSSS _{Clim}	CRPS	CRPSS _{Clim}	CRPSS _{Damp}	RPSS	Cov90	Width90
Clim	0.783	0.965	—	0.000	0.602	0.000	−0.170	0.000	0.686	1.925
Persistence	0.765	0.967	0.600	−0.003	—	—	—	—	—	—
Damp	0.668	0.840	0.514	0.243	0.515	0.145	0.000	0.181	0.713	1.800
CFS_BC	0.614	0.783	0.647	0.342	0.471	0.218	0.084	0.295	0.728	1.671
Ridge L	0.577	0.739	0.691	0.414	0.452	0.250	0.122	0.340	0.684	1.517
Ridge LG	0.477	0.612	0.780	0.598	0.370	0.385	0.281	0.471	0.724	1.341
GBM L	0.661	0.831	0.538	0.259	0.499	0.171	0.030	0.248	0.584	1.282
GBM LG	0.548	0.704	0.689	0.468	0.428	0.289	0.168	0.420	0.571	1.085
LSTM LG	0.501	0.630	0.794	0.574	0.375	0.378	0.272	0.462	0.751	1.560
Chronos	0.607	0.774	0.644	0.357	0.539	0.105	−0.047	0.099	0.310	0.578
M*	0.509	0.646	0.795	0.553	0.394	0.345	0.234	0.466	0.630	1.169
Chronos (abs.)	0.541	0.697	0.699	0.479	0.409	0.320	0.205	0.370	0.862	2.149
Chronos-Bolt	0.648	0.835	0.665	0.251	0.491	0.184	0.045	0.336	0.952	3.489
M* ₂	—	—	—	—	0.346	0.426	0.328	—	0.866	—
Hybrid	—	—	—	—	0.355	0.411	0.310	—	0.812	—

Conventions as for week 1.

Table 9

CRPSS against climatology of the main systems, with 95% intervals (blind folds).

System	W1	W2	W3–4
Damped persistence	0.134 (0.06, 0.20)	0.112 (0.04, 0.18)	0.145 (0.06, 0.21)
CFS_BC	0.415 (0.32, 0.50)	0.298 (0.21, 0.39)	0.218 (0.14, 0.28)
Ridge LG	0.297 (0.21, 0.39)	0.324 (0.23, 0.41)	0.385 (0.27, 0.48)
LSTM LG	0.273 (0.18, 0.37)	0.309 (0.20, 0.41)	0.378 (0.27, 0.48)
M*	0.295 (0.20, 0.39)	0.298 (0.20, 0.38)	0.345 (0.24, 0.43)
M* ₂ (A3)	0.323 (0.23, 0.42)	0.348 (0.24, 0.44)	0.426 (0.30, 0.52)
Hybrid (A5)	0.424 (0.34, 0.50)	0.384 (0.26, 0.50)	0.411 (0.28, 0.53)

latitude, what Monhart et al. (2018) found for bias-corrected European forecasts and He et al. (2021) and Mouatadid et al. (2023b) found for North America.

The hybrid of amendment A5 exploits that crossover (Fig. 5; Table 9). It reaches CRPSS of 0.424 (0.34–0.50), 0.384 (0.26–0.50) and 0.411 (0.28–0.53), the highest point estimates of any system at W1 and W2 and the second highest at W3–4. Against CFS_BC it is indistinguishable at W1 (ΔCRPS 0.031; $p = 0.26$) and better at W2 and W3–4 (Holm $p = 0.012$ and 0.002); against M* it is better at all three horizons (Holm $p = 0.0003$, 0.017 and 0.031). The same-rows control shows that the gain comes from the CFSv2 predictor rather than from the weekly training sample: the hybrid beats it by 0.462 and 0.262 at W1 and W2 (Holm $p = 0.0003$ and 0.012), and by a small margin at W3–4 (0.035 ; Holm $p = 0.026$, two-sided interval -0.006 to 0.070), where the CFSv2 signal itself is weak. The correlation between the bias-corrected CFSv2 anomaly and the observation falls from 0.70 at W1 to 0.21 at W3–4 on the blind folds, which explains why the hybrid converges towards its control at the longest horizon. These results are post hoc and the hybrid was not tuned; they suggest, rather than establish, that a regularised linear combination of station predictors and a dynamical forecast is the most skilful of the systems tested.

Two other hybrids behave differently. The gradient-boosting hybrid also improves on its own control at every horizon (Holm $p \leq 0.004$), but with about 185 Mondays per station its quantile models are poorly calibrated (90% coverage 0.47–0.70) and it does not beat CFS_BC. The untrained blend matches CFS_BC at W1 and improves on it at W3–4 (Holm $p = 0.001$), but it is worse than M* at W3–4, because equal weights give the weak CFSv2 forecast too much influence at that lead. A regularised model that learns how much to trust the dynamical forecast at each horizon is therefore preferable to a fixed combination, consistent with the post-processing literature (Rasp and Lerch, 2018; Slater et al., 2023).

Table 10

Hypothesis tests on the blind folds.

Hypothesis	Test	W1	W2	W3–4
H1	Clark–West, Ridge L vs Ridge LG	4.10 (Holm $< 10^{-4}$)	4.56 ($< 10^{-5}$)	4.55 ($< 10^{-5}$)
H1	Δ CRPS, Ridge L – Ridge LG	0.218 (0.016)	0.341 (0.009)	0.407 (0.009)
H1	Δ CRPS, GBM L – GBM LG	0.068 (0.264)	0.261 (0.048)	0.354 (0.004)
H2	MSSS of M* against Damp	—	—	0.409 ($p = 0.0001$)
H3	CRPSS of M* against Clim	0.295 (0.0003)	0.298 (0.0003)	0.345 (0.0003)
H4	Δ CRPS, CFS_BC – M*	–0.419 (0.999)	0.001 (0.693)	0.383 (0.012)
H5	Δ CRPS, M* – M*_2	0.098 (0.002)	0.159 (0.002)	0.241 (0.001)
H6	Δ CRPS, CFS_BC – Hybrid	0.031 (0.264)	0.274 (0.012)	0.579 (0.002)
H6	Δ CRPS, M* – Hybrid	0.450 (0.0003)	0.273 (0.017)	0.196 (0.031)
H6	Δ CRPS, control – Hybrid	0.462 (0.0003)	0.262 (0.012)	0.035 (0.026)

Δ CRPS: paired difference, reference minus system, summed over stations per issuance ($^{\circ}\text{C}$); positive values favour the system. In parentheses: one-sided p adjusted by Holm's procedure within each test over the three horizons (unadjusted for H2).

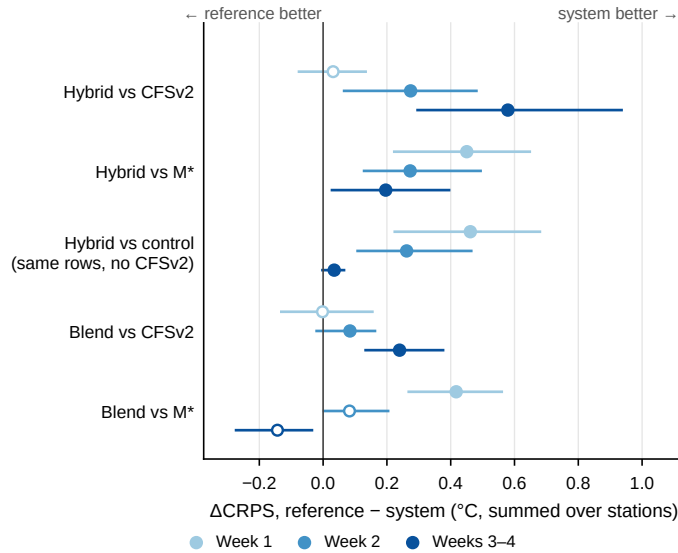


Figure 5: Paired CRPS differences of the hybrid and the blend against their references, with 95% intervals; filled markers: Holm-adjusted $p < 0.05$.

5.4. Calibration of the ensemble

M* is skilful but over-confident (Fig. 6). Its blind 90% interval covers 0.753, 0.686 and 0.630 of the outcomes, outside the pre-registered 0.85–0.95, and its PIT histogram at W3–4 is U-shaped, with 17% and 32% of the outcomes in the lowest and highest deciles. The cause is the Vincentization of members that disagree (Lichtendahl et al., 2013): averaging quantiles narrows the distribution exactly when member uncertainty is largest. The dev-selected M*_2 combines the same members with Chronos (absolute context) and a conformal calibration. Its coverage is 0.903, 0.887 and 0.866, within the target range, and its CRPS is lower than that of M* at all horizons (H5; Holm $p \leq 0.002$), with CRPSS of 0.323, 0.348 and 0.426. The selection is visible in Table 11 (Section A): the conformal calibration had the lowest cross-fitted development CRPS among the in-range configurations, and EMOS, although best on the blind CRPS (0.398 $^{\circ}\text{C}$ against 0.415 $^{\circ}\text{C}$), was not chosen because its development coverage at W3–4 (0.872) was in range but its development CRPS was higher (0.395 $^{\circ}\text{C}$ against 0.382 $^{\circ}\text{C}$). Reporting all candidates keeps the post hoc selection auditable. The hybrid, which has no calibration step, covers 0.81–0.84; CFS_BC covers 0.85 at W1 but only 0.73 at W3–4.

Hybrid subseasonal temperature forecasts at Andean stations

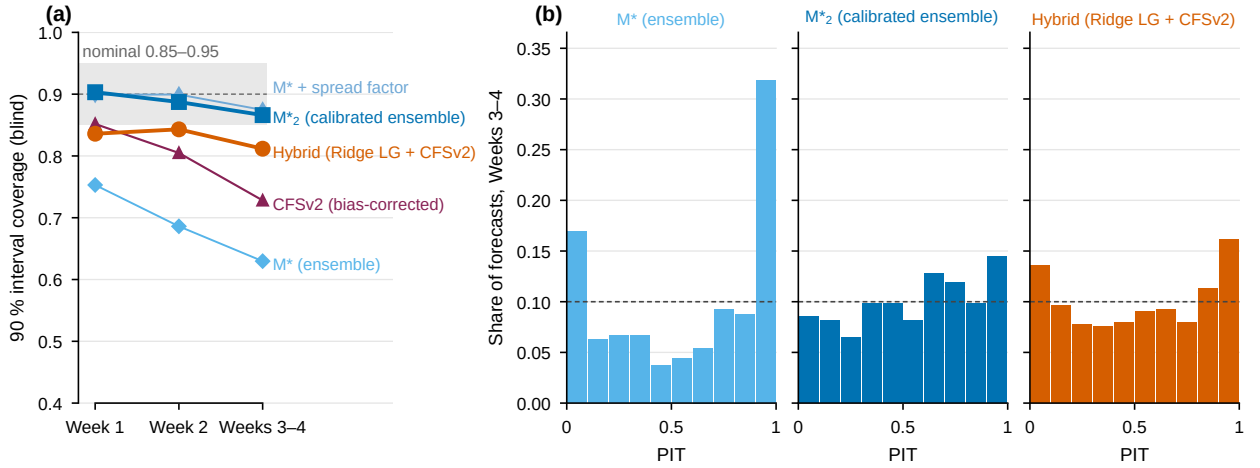


Figure 6: Calibration on the blind folds: (a) 90% interval coverage, with the nominal band; (b) PIT histograms at weeks 3–4.

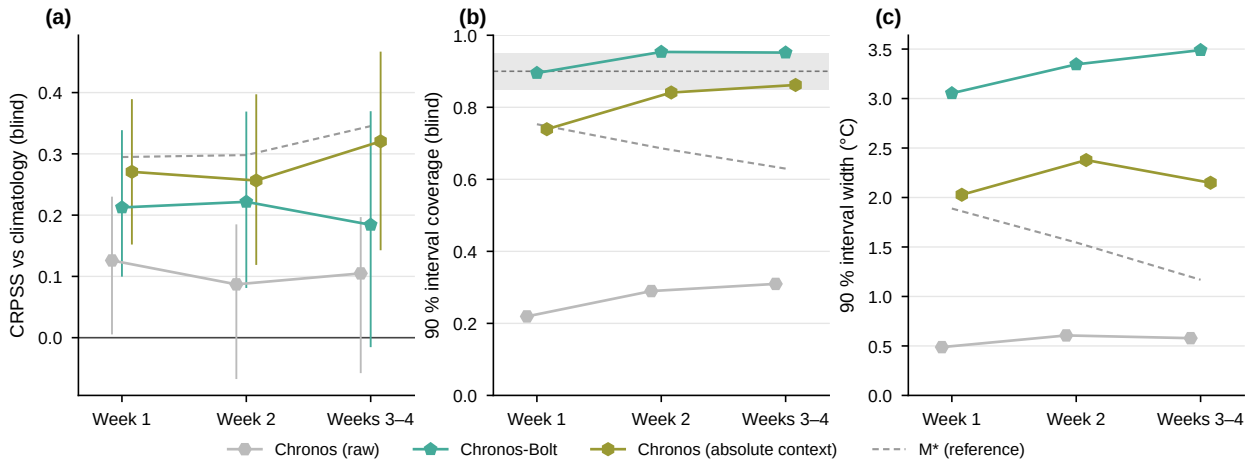


Figure 7: Chronos variants on the blind folds: (a) skill, (b) 90% interval coverage and (c) interval width, with M^* as reference.

5.5. Adapting the time-series foundation model

Raw Chronos, used zero-shot on the daily anomaly series, has the lowest probabilistic skill of the learned models (CRPSS 0.126, 0.087 and 0.105) despite an anomaly correlation (0.64–0.69) similar to that of the other models (Fig. 7; Tables 6–8). Its intervals are the problem: they cover 0.22–0.31 of the outcomes and are 0.5–0.6 °C wide, less than half the width of any other system. The diagnostic of amendment A2 locates the cause in the daily paths. Their 90% interval covers 0.21–0.22 of the daily anomalies, and their spread is about one fifth of the error of their median (path standard deviation 0.20–0.24 °C against 1.14–1.32 °C), so averaging them over a window cannot recover the missing spread. With absolute temperature as context, the model sees the seasonal cycle it was pretrained on, and its window-mean intervals widen to 2.0–2.4 °C with coverage 0.74–0.86 and CRPSS 0.271, 0.256 and 0.320, comparable to GBM LG. Chronos-Bolt, which forecasts block means directly, reaches nominal coverage (0.90–0.95) but with intervals of 3.1–3.5 °C, the widest of any system, and its CRPSS (0.18–0.22) stays below that of Chronos with absolute context. The three variants span the balance between calibration and sharpness that Gneiting et al. (2007) describe: a forecast can be calibrated by being vague, and a proper score rewards the system that is sharp subject to calibration.

Hybrid subseasonal temperature forecasts at Andean stations

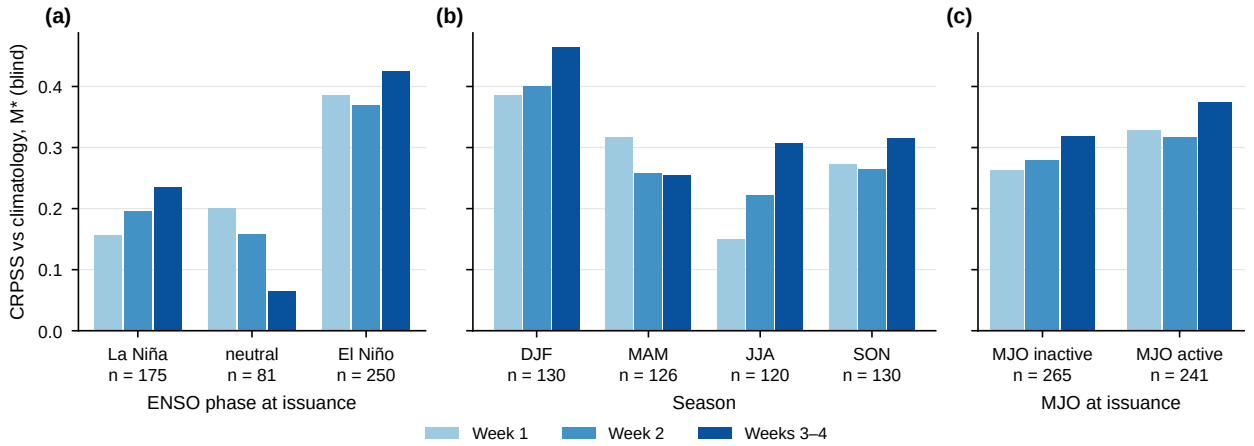


Figure 8: Skill of M* by (a) ENSO phase, (b) season and (c) MJO activity at issuance (blind folds).

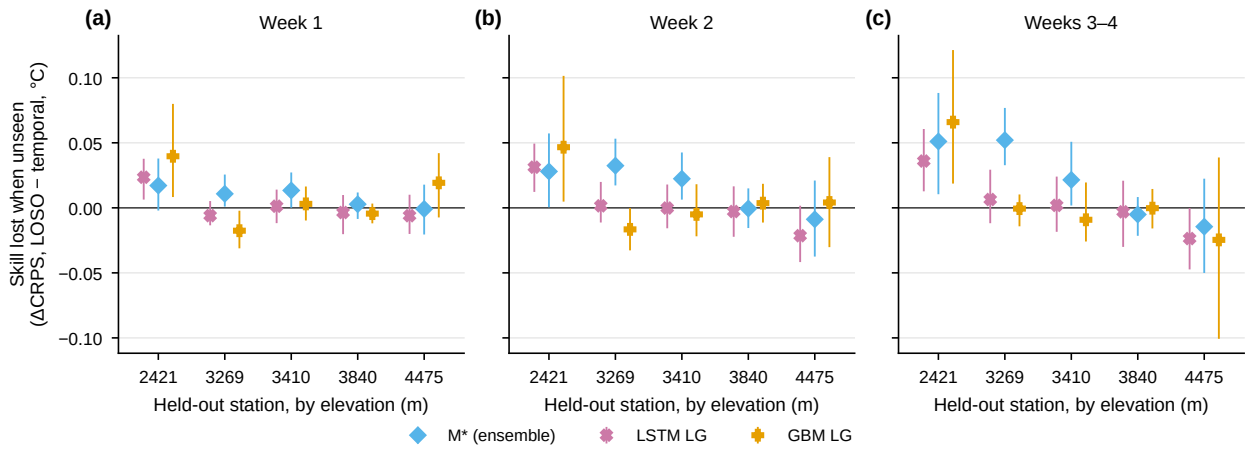


Figure 9: CRPS lost when a station is left out of training, by station elevation, with 95% intervals.

5.6. Robustness

Windows of opportunity. Skill depends on the state of ENSO at issuance (Fig. 8). The CRPSS of M* is 0.37–0.43 when El Niño conditions prevail ($n = 240$ – 250 forecasts), 0.16–0.23 under La Niña ($n = 175$) and 0.06–0.20 in neutral conditions ($n = 63$ – 81). It is also higher in austral summer (0.39–0.46 in December–February) than in winter (0.15–0.31 in June–August), and slightly higher with an active MJO (0.32–0.38) than with an inactive one (0.26–0.32). Skill is positive in every condition, so it is not confined to the 2023–24 El Niño, but that event contributes a large share of the blind forecasts; this dependence on the regime is what Mariotti et al. (2020) call a window of opportunity, and it matches the strong ENSO control of Andean temperature (Vuille et al., 2000; Garreaud, 2009). These splits are descriptive, and the neutral cell is small.

Transfer to unseen stations. When the pooled models are trained without the station they forecast, their skill changes little (Fig. 9). For M* the CRPS increases by at most 0.052 °C, and at the two highest stations it does not increase at all; the largest losses are at San José de Uzuña and Matucana at W3–4, where the CRPSS drops from 0.41 to 0.34 and from 0.44 to 0.33 (Table 14). The pooled models therefore transfer to stations within the range of the training elevations, as in the multi-basin transfer of Kratzert et al. (2019a), although five stations are too few to test transfer across climates.

Training noise and period. Re-training GBM L, GBM LG and LSTM LG with four additional seeds changes their CRPSS by a standard deviation of at most 0.005 for the gradient-boosting models and 0.020 for the LSTM, and

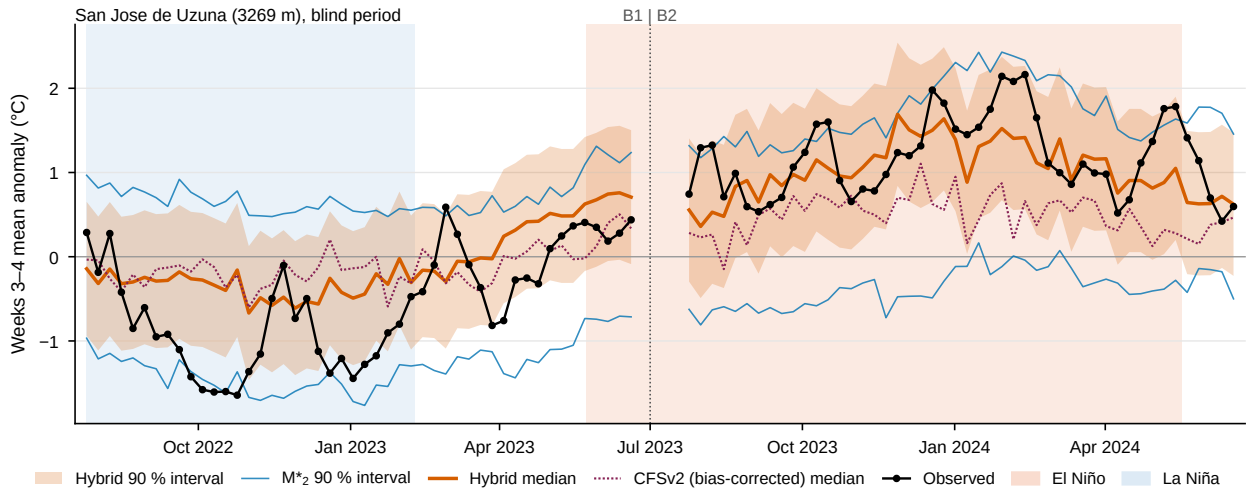


Figure 10: Weeks 3–4 forecasts at San José de Uzuña (3,269 m) over the blind period; the dotted line marks the B1–B2 fold boundary.

the ranking $GBM\ L < GBM\ LG < LSTM\ LG$ holds in every seed and horizon (Table 12; Fig. 11 in Section B). All models have higher skill on the blind folds than on the development folds (Fig. 12), M^* for example 0.29–0.35 against 0.12–0.16. The order of the models is preserved, which indicates that the increase reflects the more predictable blind years, dominated by the 2023–24 El Niño, rather than a model selection effect.

5.7. Case study

Figure 10 shows the W3–4 forecasts at San José de Uzuña, the station with the median hybrid CRPS, chosen before the figure was drawn. During the 2022–23 La Niña the observed anomalies are mostly negative; the hybrid median is negative too but weaker, and several observations fall near or below the lower edge of its 90% band. From mid-2023 the anomalies build up with the onset of El Niño, and the hybrid median follows the rise to about +1.5 °C in early 2024, above the CFSv2 median, which stays mostly between 0 and +1 °C. M^*_2 brackets the observations with a wider band. The gap before July 2023 corresponds to the issuances whose target would fall in the next fold's test year.

6. Discussion

6.1. What the system adds

The central result is that the statistical and dynamical forecasts fail in different places. CFS_BC is the best single system at W1 and the weakest among the weakest at W3–4; the station-based models are the reverse. The hybrid ridge model, which learns how much weight the CFSv2 anomaly deserves at each horizon, is at least as good as the better of the two at every horizon. This complementarity has been reported for subseasonal forecasts in North America (He et al., 2021; Mouatadid et al., 2023a) and, through bias correction of dynamical forecasts, for Europe (Monhart et al., 2018), and it is the premise of hybrid forecasting more broadly (Slater et al., 2023); our results extend it to high-elevation stations in the tropics, where ENSO is the dominant source of predictability (Vuille et al., 2000; Lavado-Casimiro and Espinoza, 2014; Sulca et al., 2017). In engineering terms, the hybrid is a low-cost addition to an existing station-based system: it requires one dynamical forecast per week and per station and a linear model refitted on a few hundred cases.

6.2. Lessons for time-series foundation models

Chronos is a capable point forecaster on this problem but, used as offered, a poor probabilistic one. The problem is not the accuracy of its median but the spread of its sample paths, which is too small at the daily scale to survive averaging. Giving the model the absolute temperature, closer to the data it was pretrained on (Ansari et al., 2024), restores most of the spread, and Chronos-Bolt shows the opposite failure. For users of foundation models in environmental applications, the practical implication is that interval coverage of the derived quantity, not only point

accuracy, has to be checked before such models enter an ensemble, and that the input representation matters as much as the model choice (Liang et al., 2024; Tan et al., 2024).

6.3. Simple versus complex models

Ridge LG matches or exceeds every more complex model, including the LSTM, the foundation models and the ensemble, on the blind folds. The gain from adding the large-scale indices (H1) is larger than any gain from changing the architecture, and the best gradient-boosting and LSTM models differ by less than their seed-to-seed range at W1. With about ten years of data at five stations, the limiting factor is the information in the predictors rather than the capacity of the model, a conclusion that echoes forecasting competitions (Petropoulos et al., 2022) and contrasts with settings where large training sets let deep models learn shared structure across hundreds of sites (Kratzert et al., 2019b).

6.4. Operational implications

A forecasting service needs probabilities it can trust. The conformal calibration selected on development years delivers 90% intervals that cover close to 90% of the blind outcomes without sacrificing skill, and the case study shows forecasts that track an evolving El Niño anomaly several weeks in advance. Both are relevant for frost and agricultural planning, where the cost of a missed cold anomaly differs from that of a false alarm. The secondary results support this use: Ridge LG has positive skill for the minimum temperature at every horizon (CRPSS 0.21–0.28) and for the probability of frost (Brier skill 0.13–0.29; Table 13), although the frost index is computed from hourly means and understates true frost.

6.5. Limitations

The study has five limitations that bound its conclusions. (i) Five stations and two blind years give wide confidence intervals; several differences between the best systems are not distinguishable, and the results should not be generalised to the whole Andes without further stations. (ii) The hybrid, M^*_2 and the Chronos variants are post hoc; their designs were fixed before their own scores but after the blind scores of the pre-registered systems, and they should be confirmed on later years. (iii) The hybrid is trained on about 185 Mondays per station because the open CFSv2 archive starts in 2018, and its 90% coverage (0.81–0.84) is below the target range; a calibration step could not be selected without development years. (iv) The only dynamical reference is CFSv2 at 1° resolution; the ECMWF extended-range system, usually the strongest S2S reference, was not available to us, so our comparison is with an open dynamical forecast rather than with the state of the art. (v) The blind period includes the 2023–24 El Niño, which raises the skill of all systems; the conditional results show positive skill in all ENSO phases, but the size of the gains is tied to that period.

7. Conclusions

Subseasonal temperature forecasts at high-Andean stations can be both skilful and calibrated, provided that the forecasting system is designed and evaluated with the same discipline. That is the thread that connects the methodological and technical results of this study.

On the methodological side, freezing the models and their hyperparameters before scoring two blind years, separating training and test targets by an embargo and testing every comparison with a block bootstrap and Holm's correction made it possible to tell real skill from favourable sampling. The same protocol exposed what cross-validation alone would have hidden: the blind years were more predictable than the development years for every model, and the primary ensemble, although skilful, was over-confident. A leakage-free anomaly decomposition gave statistical, deep and foundation models a common target, and declaring every later analysis as a dated amendment kept the post hoc results distinguishable from the confirmatory ones, in line with recommendations for forecast evaluation (Tashman, 2000; Bergmeir and Benítez, 2012; DelSole and Tippet, 2014).

On the technical side, four findings follow. A ridge regression that adds the bias-corrected CFSv2 forecast to station and large-scale predictors matched CFSv2 at week 1 and improved on it at weeks 2–4, with skill scores of 0.42, 0.38 and 0.41 against climatology; its same-rows control attributes the gain to the dynamical predictor, extending to the tropical Andes the complementarity reported elsewhere (Monhart et al., 2018; He et al., 2021). The ENSO and MJO indices contributed more skill than any change of architecture, so that a regularised linear model was as good as a recurrent network and a foundation model (Petropoulos et al., 2022). Chronos, used zero-shot, produced intervals about four times too narrow for window-mean anomalies until it received absolute-temperature context, a reminder that foundation models must be checked for calibration on the quantity a user needs (Gneiting et al., 2007; Ansari

et al., 2024). Finally, a conformal calibration chosen on development years turned an over-confident ensemble into one whose 90% intervals covered close to 90% of the blind outcomes (Romano et al., 2019).

The two sides depend on each other. The technical results are credible because the protocol was fixed in advance, and the protocol is worth its cost because the results it validates change what a forecaster can offer for frost and agricultural decisions. The evidence comes from five stations and two blind years, and the hybrid and the calibrated ensemble were defined after the first blind scores; confirming them on later years and more stations, adding the ECMWF extended-range forecasts and calibrating the hybrid are the next steps. Within these limits, the study shows that combining station-based learning with an open dynamical forecast yields calibrated guidance for the weeks that matter most to Andean farmers and water managers.

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A. Supplementary tables

B. Supplementary figures

Table 11Selection of M^*_2 among 16 configurations (amendment A3).

Members	Combination	Calibration	Dev CRPS	Dev Cov90 (W1/W2/W3-4)	Blind CRPS
With Chronos	Vincentization	None	0.385	0.81 / 0.84 / 0.83	0.419
With Chronos	Vincentization	Spread factor	0.383	0.89 / 0.90 / 0.90	0.415
With Chronos	Vincentization	Conformal	0.382	0.89 / 0.90 / 0.90	0.415
With Chronos	Vincentization	EMOS	0.395	0.90 / 0.89 / 0.87	0.398
With Chronos	Linear pool	None	0.384	0.83 / 0.86 / 0.88	0.420
With Chronos	Linear pool	Spread factor	0.383	0.89 / 0.90 / 0.90	0.418
With Chronos	Linear pool	Conformal	0.383	0.89 / 0.90 / 0.90	0.418
With Chronos	Linear pool	EMOS	0.395	0.90 / 0.89 / 0.87	0.398
Without Chronos	Vincentization	None	0.394	0.81 / 0.79 / 0.76	0.444
Without Chronos	Vincentization	Spread factor	0.390	0.89 / 0.88 / 0.90	0.436
Without Chronos	Vincentization	Conformal	0.388	0.89 / 0.89 / 0.90	0.436
Without Chronos	Vincentization	EMOS	0.400	0.90 / 0.89 / 0.89	0.423
Without Chronos	Linear pool	None	0.393	0.81 / 0.80 / 0.78	0.443
Without Chronos	Linear pool	Spread factor	0.390	0.89 / 0.89 / 0.89	0.436
Without Chronos	Linear pool	Conformal	0.388	0.90 / 0.89 / 0.90	0.436
Without Chronos	Linear pool	EMOS	0.400	0.90 / 0.89 / 0.89	0.423

Development CRPS and coverage are cross-fitted over D1–D3. The selected configuration (bold) has the lowest development CRPS among those with coverage in 0.85–0.95 at every horizon.

Table 12

CRPSS against climatology over five training seeds (blind folds).

Model	Horizon	Mean	SD	Min	Max
GBM L	W1	0.222	0.001	0.220	0.223
GBM L	W2	0.159	0.004	0.155	0.164
GBM L	W3–4	0.178	0.005	0.171	0.184
GBM LG	W1	0.239	0.002	0.236	0.241
GBM LG	W2	0.236	0.002	0.234	0.239
GBM LG	W3–4	0.289	0.004	0.283	0.293
LSTM LG	W1	0.262	0.010	0.248	0.273
LSTM LG	W2	0.296	0.020	0.266	0.320
LSTM LG	W3–4	0.363	0.020	0.341	0.389

Table 13

Skill for the secondary targets (blind folds).

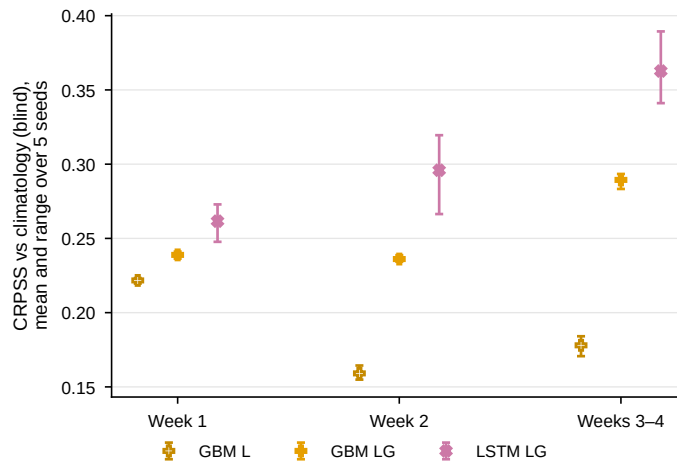
Target	Model	CRPSS W1	CRPSS W2	CRPSS W3–4	BSS W1	BSS W2	BSS W3–4
TT_min	Damp	0.090	0.067	0.103	−0.025	−0.072	0.042
TT_min	GBM LG	0.123	0.124	0.149	0.021	0.070	0.130
TT_min	Ridge LG	0.205	0.225	0.282	0.134	0.265	0.291
TT_max	Damp	0.064	0.061	0.084	—	—	—
TT_max	GBM LG	0.191	0.160	0.160	—	—	—
TT_max	Ridge LG	0.215	0.192	0.218	—	—	—

CRPSS against climatology of the daily minimum (TT_min) and maximum (TT_max) temperature anomalies. BSS: Brier skill score of the frost probability (TT_min below 0 °C, from hourly means).

Table 14

Leave-one-station-out transfer of M* (blind folds).

Station (elevation)	Horizon	CRPSS temporal	CRPSS LOSO	Δ CRPS (95% interval)
Matucana (2,421 m)	W1	0.366	0.334	0.017 (−0.002, 0.038)
Matucana (2,421 m)	W2	0.361	0.304	0.028 (0.000, 0.057)
Matucana (2,421 m)	W3–4	0.442	0.333	0.051 (0.010, 0.088)
San José de Uzuña (3,269 m)	W1	0.336	0.323	0.011 (0.001, 0.026)
San José de Uzuña (3,269 m)	W2	0.355	0.311	0.032 (0.017, 0.053)
San José de Uzuña (3,269 m)	W3–4	0.410	0.338	0.052 (0.033, 0.077)
Candarave (3,410 m)	W1	0.303	0.287	0.013 (0.000, 0.027)
Candarave (3,410 m)	W2	0.314	0.285	0.022 (0.006, 0.043)
Candarave (3,410 m)	W3–4	0.348	0.320	0.021 (0.002, 0.051)
Carania (3,840 m)	W1	0.097	0.091	0.003 (−0.009, 0.012)
Carania (3,840 m)	W2	0.067	0.068	−0.001 (−0.016, 0.015)
Carania (3,840 m)	W3–4	0.128	0.147	−0.005 (−0.022, 0.008)
Imata (4,475 m)	W1	0.305	0.306	−0.001 (−0.020, 0.018)
Imata (4,475 m)	W2	0.293	0.304	−0.009 (−0.037, 0.021)
Imata (4,475 m)	W3–4	0.299	0.317	−0.014 (−0.050, 0.022)

CRPSS with the station in training (temporal) and left out (LOSO); Δ CRPS: increase in CRPS (°C) with its 95% interval.**Figure 11:** Seed variability of CRPSS: mean and range over five seeds.

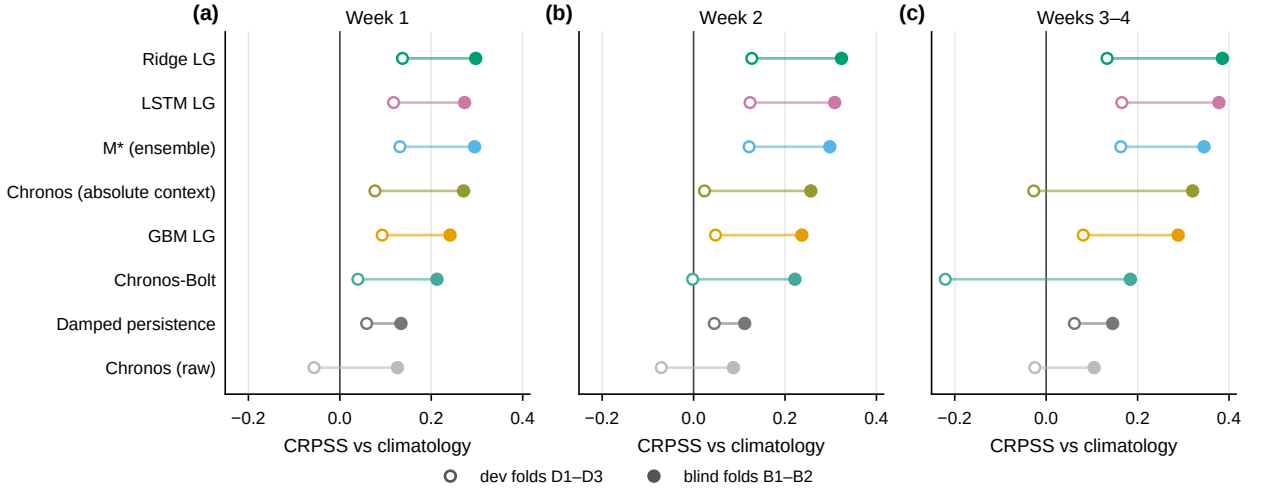


Figure 12: Skill on the development (hollow) and blind (filled) folds, by model and horizon.

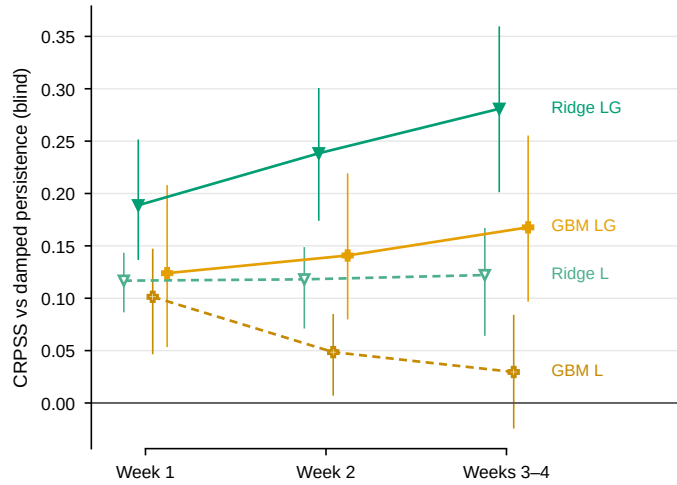


Figure 13: Skill against damped persistence of local-only (L) and local plus large-scale (LG) models.